COURSE CONTENT SUMMARY — FOR TRANSFER EVALUATION

CHEM&121: Introduction to Chemistry

Fall 2026 (September 21 – December 10, 2026)

What's Covered

Everything covered in this course, organized by topic and week — useful for comparing against an equivalent course at another institution, and for previewing what's ahead before the term starts.

Built from the Fall 2026 Chemfolio schedule — first term using this material, so some groupings may shift slightly as the term settles in.

Week 1: Atoms, Isotopes, and Electron Configuration

What this week covers:

- Elements, atomic structure, and isotopes (protons/neutrons)
- The periodic table; electronic structure and configuration (electrons)
- Valence electrons and periodic trends
- Lab: Safety orientation; significant figures applied to measurement and calculation; density of an unknown; dimensional analysis
- Required Reading: Smith Ch. 2 (2.1–2.8)

▼ **Weekly logistics**

- Activity: A01 (2.1–2.3)
- Activity: A02 (2.4–2.8)
- Lab 1

Week 2: Ionic and Covalent Bonding

What this week covers:

- Introduction to bonding; ions; ionic compounds and their naming
- Physical properties of ionic compounds; polyatomic ions
- Covalent bonding; Lewis structures; naming covalent compounds
- Lab: Density, Seawater, and Thermohaline Circulation
- Required Reading: Smith Ch. 3 (3.1–3.9)
- Structure guide: Naming Compounds, Lewis Structures

▼ **Weekly logistics**

- Activity: A03 (3.1–3.6)
- Activity: A04 (3.7–3.9)

- Lab 2
- Also this week: Unit Quiz 1 (Ch. 1 + Ch. 2, no dimensional analysis); Writing Workshop 1

Week 3: Molecular Shape, Polarity, and Intermolecular Forces

What this week covers:

- Molecular shape; electronegativity and bond polarity; polarity of molecules
- Energy; the three states of matter; intermolecular forces; boiling and melting point
- Lab: Molecular Shapes and GHGs
- Writing Workshop 2
- Required Reading: Smith Ch. 3 (3.10–3.12), Ch. 4 (4.1–4.4)
- Structure guide: Shape, Polarity, and Intermolecular Forces

▼ Weekly logistics

- Activity: A05 (3.10–3.12)
- Activity: A06 (4.1–4.4)
- Lab 3 (continued into simulations alongside Writing Workshop 2)

Week 4: Heat, Phase Changes, and the Mole

What this week covers:

- Specific heat; energy and phase changes; heating and cooling curves
- Mass balance; introduction to reactions, balancing equations, reaction types, and oxidation/reduction
- Lab: Copper Cycle and AMD
- Library Visit 1
- Required Reading: Smith Ch. 4 (4.5–4.7), Ch. 5 (5.1–5.4)

▼ Weekly logistics

- Activity: A07 (4.5–4.7)
- Activity: A08 (5.1–5.4)
- Lab 4 (continued — weighing copper — alongside Library Visit 1)

Week 5: Stoichiometry and Reaction Energy

What this week covers:

- The mole and Avogadro's number; mass-to-mole conversions; mole and mass calculations in chemical equations
- Energy changes in reactions; reaction rates
- Lab: Urban Gardens (soil sampling)
- Required Reading: Smith Ch. 5 (5.5–5.10)

▼ Weekly logistics

- Activity: A09 (5.5–5.8)
- Activity: A10 (5.9–5.10)

- Lab 5 (finished with a Graphing Workshop)
- Also this week: Unit Quiz 2 (Ch. 3 + Ch. 4)

Week 6: Equilibrium, Pressure, and Kinetic Molecular Theory

What this week covers:

- Equilibrium; Le Châtelier's principle; body temperature regulation
- Gases and pressure; Boyle's, Charles's, and Gay-Lussac's laws
- Lab: Soil Quality Analysis (Day 1) and Water Quality Analysis (Day 2)
- Library Visit 2
- Required Reading: Smith Ch. 5 (5.11–5.12), Ch. 6 (6.1–6.4)

▼ **Weekly logistics**

- Activity: A11 (5.11–5.12)
- Activity: A12 (6.1–6.4)
- Lab 6 (two-day lab: NOx tube deployment Day 1, soil testing completed Day 2 alongside Library Visit 2)

Week 7: Gas Laws and Air Quality

What this week covers:

- The combined gas law; Avogadro's law; the ideal gas law; Dalton's law and partial pressures; ozone and CO2 in the atmosphere
- Solubility fundamentals; electrolytes and nonelectrolytes
- Lab: Air Pollution and Pollinators
- Required Reading: Smith Ch. 6 (6.5–6.9), Ch. 7 (7.1–7.4)

▼ **Weekly logistics**

- Activity: A13 (6.5–6.9)
- Activity: A14 (7.1–7.4)
- Lab 7 (NOx tube collection, ozone strip placement; finished with a Graphing Workshop for air-quality results)
- Also this week: Unit Quiz 3 (Ch. 5)

Week 8: Concentration, Acids, and Bases

What this week covers:

- Percent concentration; molarity; dilution; osmosis and dialysis
- Introduction to acids and bases; Brønsted–Lowry reactions; acid/base strength
- Lab: Drinking Water
- Required Reading: Smith Ch. 7 (7.5–7.8), Ch. 8 (8.1–8.3)

▼ **Weekly logistics**

- Activity: A15 (7.5–7.8)
- Activity: A16 (8.1–8.3)
- Lab 8 (Analysis — "My Waterway Report")

- Calendar note: 11/11 campus holiday — lecture and lab still held this week

Week 9: pH, pOH, and Buffers

What this week covers:

- Dissociation of water; the pH scale; common acid-base reactions
- Buffers; buffers in the blood
- Lab: Surface and Ground Water
- Required Reading: Smith Ch. 8 (8.4–8.6, 8.8–8.9)

▼ **Weekly logistics**

- Activity: A17 (8.4–8.6)
- Activity: A18 (8.8–8.9)
- Lab 9 (finished with Water Quality Index calculations)
- Also this week: Unit Quiz 4 (Ch. 6 + Ch. 7)

Week 10: Nuclear Energy, Medical Applications, and Final Project

What this week covers:

- Isotopes and radioactivity; nuclear reactions; half-life
- Detecting and measuring radioactivity; medical uses of radioisotopes
- Nuclear fission and fusion; medical imaging without radioactivity
- Lab: UV Radiation and Alginate Beads
- Final Project: Teach Your Community (TYC) — final exam assessment, 10% of course grade, due Thursday, 12/10
- Required Reading: Smith Ch. 9 (9.1–9.7)

▼ **Weekly logistics**

- Activity: A19 (9.1–9.3)
- Activity: A20 (9.4–9.7)
- Lab 10
- Field trip / TYC Outline work sessions
- Finals week: Unit Quiz 5 (Ch. 8 + Ch. 9); course potluck
- Calendar note: campus closed 11/25–11/26 during this content window